

Phishing Threat & Audience Analysis – Hocking College

1. Phishing Attack Types

Email Phishing

Email phishing involves attackers sending messages that appear to come from trusted sources such as the college IT department or professors. These emails often include links to fake login pages designed to steal credentials. The attacker's goal is typically to gain access to student or staff accounts for further exploitation, such as accessing sensitive data or sending additional phishing emails from compromised accounts.

Smishing (SMS Phishing)

Smishing uses text messages to trick users into clicking malicious links or responding with sensitive information. Messages may claim urgent issues like account suspension or financial aid problems. Attackers aim to exploit the immediacy of SMS communication to bypass user skepticism and quickly capture login credentials or personal information.

QR Phishing (Quishing)

QR phishing involves malicious QR codes placed on posters, emails, or digital platforms that redirect users to fraudulent websites. These attacks are effective in environments like college campuses where QR codes are commonly used for events and resources. The attacker's objective is to bypass traditional link scrutiny by hiding malicious URLs behind QR codes.

Credential Harvesting

Credential harvesting is often embedded within other phishing methods and involves directing users to fake login portals that closely resemble legitimate systems like student email or learning platforms. The attacker collects usernames and passwords, which can then be used for account takeover or unauthorized access to institutional systems.

MFA Fatigue / Push Attacks

In MFA fatigue attacks, attackers repeatedly send authentication requests to a user's device hoping they will approve one out of annoyance or confusion. This method relies on user fatigue and lack of awareness. Once approved, attackers gain access to secured accounts despite multi-factor authentication protections.

2. Human Factors Analysis

Phishing attacks are successful because they exploit predictable human behaviors and emotions rather than technical vulnerabilities.

Urgency

Attackers create a sense of urgency to pressure users into acting quickly without thinking. For example, an email stating “Your account will be locked in 30 minutes” can cause users to click links without verifying legitimacy.

Authority

Messages often appear to come from trusted figures such as professors, IT staff, or administrators. A student is more likely to trust an email that appears to be from “Hocking College IT Support” requesting a password reset.

Fear

Fear-based messages threaten consequences such as losing access to classes, financial aid, or accounts. This emotional response can override critical thinking and lead users to comply with malicious requests.

Curiosity

Attackers use curiosity by offering incentives or intriguing subject lines like “Updated grades available” or “Important campus announcement.” This encourages users to click links without verifying the source.

3. Target Audiences at Hocking College

Students

Students are highly vulnerable due to frequent email use, reliance on online platforms, and limited cybersecurity awareness. Many students quickly respond to messages related to grades, financial aid, or account access without verifying legitimacy. They are most likely to encounter email phishing and smishing attacks that mimic campus services.

Faculty

Faculty members are targets due to their access to student records, grading systems, and institutional resources. They often receive high volumes of email, making it easier for phishing messages to blend in. Faculty may be more susceptible to credential harvesting attacks disguised as system updates or administrative requests.

Staff

Administrative staff are particularly vulnerable because they handle sensitive data such as financial records, student information, and internal communications. Their roles often require quick responses to emails, increasing the likelihood of falling for urgency-based attacks. They are common targets for credential harvesting and MFA fatigue attacks aimed at gaining deeper system access.

4. Potential Impact

Phishing attacks can have significant consequences for Hocking College.

Data Exposure

Compromised accounts can lead to unauthorized access to student records, personal information, and institutional data, potentially violating privacy regulations.

Financial Impact

Attackers may redirect payments, commit financial fraud, or exploit compromised accounts to request unauthorized transactions.

Account Compromise

Stolen credentials allow attackers to access email systems, learning platforms, and internal tools, enabling further phishing attacks within the organization.

Institutional Risk

Successful attacks can damage the college's reputation, reduce trust among students and staff, and potentially lead to legal or compliance issues.